

Toluwani Samuel Aremu

AI Safety | Trustworthy AI | Responsible AI
[Certifications](#) • [Email](#) • [GitHub](#) • [Google Scholar](#) • [LinkedIn](#) • [Website](#)

RESEARCH INTERESTS

- **Risks:** [Threat Modeling](#) | [Failure Analysis](#) | [Long-term/Existential Risks](#)
- **Safety:** [Scalable Oversight](#) | [Evaluation](#) | [Debate](#) | [Cooperative AI](#) | [Alignment Theories](#)
- **Society:** [AI for Good](#) | [Impacts of AI development](#) | [Ethical Human-AI Interaction](#) | [AI Governance and Policy](#)
- **Security:** [Watermarking for Provenance and Attribution](#) | [Adversarial Attacks and Defences](#) | [Monitoring Techniques against Jailbreaks](#)

EDUCATION

- Machine Learning (PhD) | **Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI), UAE** AUG 2023 – PRESENT
 - **Advisor:** [Nils Lukas](#).
 - **Thesis:** [Robust Watermarking and Monitoring for Trustworthy Generative AI Systems](#).
 - **Expected Completion Date:** [May 2027](#).
- Machine Learning (MSc) | **Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI), UAE** JAN 2021 – JAN 2023
- Computer Science (MSc) | **University of Ibadan (UI), Nigeria** MAY 2018 – MAY 2020
- Computer Science (BSc) | **Adeleke University, Nigeria** OCT 2012 – JUL 2016
- Theory and Practice of Music (Diploma) | **Associated Board of the Royal Schools of Music (ABRSM), UK** AUG 2012 – AUG 2015

RESEARCH EXPERIENCE

- Visiting Researcher | **Center for Frontier AI Research (CFAR), A*STAR** APR 2026 – PRESENT
 - **Research Areas:** [Watermarking for Generative AI](#).
 - **Host:** [Jie Zhang](#).
- Doctoral Researcher | [Spot-AI Lab](#), MBZUAI AUG 2023 – PRESENT
 - **Research Areas:** [Watermarking for Generative AI](#), [Alignment](#), [AI Governance](#).
- Research Assistant | [MCR Lab](#), MBZUAI FEB 2023 – AUG 2023
 - **Research Areas:** [Generative AI Applications](#).
- Graduate Research Assistant | [SPriNT-AI Lab](#), MBZUAI JAN 2021 – JAN 2023
 - **Research Areas:** [Privacy-Preserving ML](#).

WORK EXPERIENCE

- Technical Co-Founder/Lead Quant | **Stealth (Incubated under [Kora](#))** FEB 2026 – PRESENT
- AI Platform Developer (Contract) | **Vadis Media, Switzerland** MAY 2025 – JUL 2025
- Applied Scientist (Intern) | **M42, UAE** FEB 2023 – APR 2023
- Data Analyst | **State Government of Osun, Nigeria** JAN 2018 – DEC 2018
- IT Support Engineer | **Government of Ekiti State, Nigeria** NOV 2016 – NOV 2017
- Hardware Engineer (Intern) | **Lautech Teaching Hospital, Nigeria** JUN 2015 – NOV 2015
- Network Engineer (Intern) | **OAU-INTECU, Nigeria** JUN 2014 – NOV 2014

PUBLICATIONS (* denotes equal contribution, ** denotes working paper, 🏆 denotes Spotlight, AR denotes acceptance rate)

- **[**] Aremu, Toluwani**, Nils Lukas, and Jie Zhang. "Watermarking Should Be Treated as a Monitoring Primitive." arXiv preprint arXiv:2605.13095 (2026).
- **[**]** Toyin, Hawau Olamide, Mutiah Apampa, **Toluwani Aremu**, Humaid Alblooshi, Ana Rita Valente, Gonçalo Leal, Zhengjun Yue, Zeerak Talat, and Hanan Aldarmaki. "Aligning Stuttered-Speech Research with End-User Needs: Scoping Review, Survey, and Guidelines." arXiv preprint arXiv:2604.20535 (2026).
- **[**] Aremu, Toluwani**, Daniil Ognev, Samuele Poppi, and Nils Lukas. "Robust Safety Monitoring of Language Models via Activation Watermarking." arXiv preprint arXiv:2603.23171 (2026).
- **[CVPR-AdvML]** Fares, Samar*, Klea Ziu*, **Toluwani Aremu***, Nikita Durasov, Martin Takáč, Pascal Fua, Karthik Nandakumar, and Ivan Laptev. "Mirrorcheck: Efficient adversarial defense for vision-language models." The 6th Workshop of Adversarial Machine Learning on Computer Vision: Safety of Vision-Language Agents. 2026.
- **[**] Aremu, Toluwani**, Noor Hussein, Munachiso Nwadike, Samuele Poppi, Jie Zhang, Karthik Nandakumar, Neil Gong, and Nils Lukas. "Mitigating Watermark Forgery in Generative Models via Randomized Key Selection." arXiv preprint arXiv:2507.07871 (2025).
- **[ICML 🏆 (Top 2.6%), AR: 26.9%]** Diaa, Abdulrahman, **Toluwani Aremu**, and Nils Lukas. "Optimizing Adaptive Attacks against Watermarks for Language Models." In International Conference on Machine Learning, pp. 13546-13569. PMLR, 2025.
- **[Patent]** KANG, Wong Yu, **Toluwani AREMU**, Younes BALAH, Abdulmoteleb El Saddik, Maryam NADEEM, and Ivo Gollini NAVARRETE. "System and method for scanning faces and discovering scholarly information." U.S. Patent 12,423,369, issued September 23, 2025.
- **[AI Magazine, AR: 24%]** **Aremu, Toluwani**, Oluwakemi Akinwehinmi, Chukwuemeka Nwagu, Syed Ishtiaque Ahmed, Rita Orji, Pedro Arnau Del Amo, and Abdulmoteleb El Saddik. "On the reliability of Large Language Models to misinformed and demographically informed prompts." Association for the Advancement of Artificial Intelligence (AAAI) AI Magazine 46, no. 1 (2025): e12208.
- **[ACL (Main), AR: 20.3%]** Nwadike, Munachiso S., Zangir Iklasov, **Toluwani Aremu**, Tatsuya Hiraoka, Benjamin Heinzerling, Velibor Bojkovic, Hilal AlQuabeh, Martin Takáč, and Kentaro Inui. "Library-Like Behavior In Language Models is Enhanced by Self-Referencing Causal Cycles." In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp. 25365-25377. 2025.
- **[CVPR 🏆 (Top 13.5%), AR: 22.1%]** Vayani, Ashmal, Dinura Dissanayake, Hasindri Watawana, Noor Ahsan, Nevasini Sasikumar,..., **Toluwani Aremu** et al. "All languages matter: Evaluating Imms on culturally diverse 100 languages." In Proceedings of the Computer Vision and Pattern Recognition Conference, pp. 19565-19575. 2025.

- [\[IJCAI, AR: 14%\]](#) Tastan, Nurbek, Samar Fares, **Toluwani Aremu**, Samuel Horvath, and Karthik Nandakumar. "Redefining contributions: shapley-driven federated learning." In Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, pp. 5009-5017. 2024.
- [\[SAI, AR: 37%\]](#) **Aremu, Toluwani**, Li Zhiyuan, Reem Alameeri, Mustaqeem Khan, and Abdulmoteleb El Saddik. "SSIVD-net: a novel salient super image classification and detection technique for weaponized violence." In Science and Information Conference, pp. 16-35. Cham: Springer Nature Switzerland, 2024.
- [\[Preprint\]](#) **Aremu, Toluwani**. "Unlocking Pandora's Box: Unveiling the Elusive Realm of AI Text Detection." Available at SSRN 4470719 (2023).
- [\[SaTML, AR: 26.3%\]](#) **Aremu, Toluwani**, and Karthik Nandakumar. "Polykervnets: Activation-free neural networks for efficient private inference." In 2023 IEEE Conference on Secure and Trustworthy Machine Learning (SaTML), pp. 593-604. IEEE, 2023.

OTHER EXPERIENCE

- **Projects**
 - Accented Speech Recognition | **Course Project (MBZUAI)** AUG 2021 – DEC 2021
 - Racial Bias Mitigation in Self-Supervised Face Recognition Architecture | **Course Project (MBZUAI)** JAN 2021 – MAY 2021
 - Gender Bias Mitigation in Word Embeddings | **Course Project (MBZUAI)** JAN 2021 – MAY 2021
- **Teaching**
 - Teaching Assistant, Introduction to AI (AI1010) | **MBZUAI** AUG 2025 – DEC 2025
 - Teaching Assistant, Mathematical Foundations of AI (MTH701) | **MBZUAI** AUG 2024 – DEC 2024
 - Teaching Assistant, Object Oriented Programming (OOP) | **University of Ibadan** JAN 2019 – MAY 2019
- **Talks/Presentations**
 - Mitigating Watermark Forgery Attacks in Generative Models via Multi-Key Watermarking | **MBZUAI Research Showcase, UAE** OCT 2025
 - Literature Review and Research Methodologies | **LyngualLabs, Nigeria** MAY 2025
 - Optimizing Adaptive Attacks against Content Watermarks for Language Models | **ICLR-WMARK Workshop, Singapore** APR 2025
 - PolyKervNets: Activation-Free Neural Networks for Efficient Private Inference | **SaTML, USA** FEB 2023
 - Ethical Perspectives of AI | **Department of Material Sciences, University of Denver, USA** JUL 2022
- **Service (Reviewer | Area Chair | Program Chair | Organizer | Editor)**
 - **Conferences:**
 - [CVPR](#) 2026
 - [NeurIPS](#) 2025, 2026
 - [ICLR](#) 2025, 2026
 - [ICML](#) 2025, 2026
 - [AAAI](#) 2025, 2026
 - [AISTATS](#) 2025
 - [DLI](#) 2025
 - **Workshops:**
 - [AI4GOOD@ICML](#) 2026
 - [HRAIM](#) 2025
 - [SafeGenAI@NeurIPS](#) 2025
 - [WMark@ICLR](#) 2025
 - **Journals:**
 - [IEEE TPAMI](#) 2025
 - [IEEE Access](#) 2025
 - [AI Magazine](#) 2025
 - [OSJ](#) 2025
 - [CHBAH](#) 2025
- **Leadership/Volunteering**
 - Resident Life Assistant | **MBZUAI Undergraduate Student Life** AUG 2025 – MAY 2027
 - Volunteering Member | **EVOLVE-MBZUAI** AUG 2024 – MAY 2027
 - Volunteering Member | **Spruce Up Your Space** AUG 2024 – MAY 2027
 - Associate Editor | **ML Department Research Blog** JAN 2024 – MAY 2027
 - Student Editor | **MBZUAI Research Blog** JAN 2024 – MAY 2027
 - Mentor | **Research Mentorship for African Research Enthusiasts** JAN 2021 – PRESENT

HONORS & AWARDS

- ICML'26 **Gold** Reviewer Award (Top 25%) MAY 2026
- MBZUAI Award of Recognition for Service and Contribution to the MBZUAI Alumni Body FEB 2026
- MBZUAI ICML Conference Travel Grant | **Value of Award: \$4,500** JUL 2025
- MBZUAI ICLR Conference Travel Grant | **Value of Award: \$4,500** APR 2025
- MBZUAI PhD Fully Funded Fellowship | **Value of Award: ~\$400,000** AUG 2023 - MAY 2027
- MBZUAI MSc Fully Funded Fellowship | **Value of Award: ~\$175,000** JAN 2021 - JAN 2023
- UAE **Golden** Visa for Talented Persons/Specialists in Science OCT 2022
- MBZUAI Award of Appreciation for Iconic Representation and Student Hospitality AUG 2022
- ProjectSet Innovation Challenge for Entrepreneurship (ICE-22) | **Value of Award: ~\$3,000 in Cloud Credits** MAY 2022
- Top 100, DeepLearning.AI Data-Centric AI Competition AUG 2021
- NYSC-FRSC Award for the Most Creative Corp Member OCT 2017
- 2015 AUE-NACOSS Award for the Best Student Programmer JUL 2015

REFERENCES

- Available on request